

Generation Interconnection System Impact Study Report

For Queue Project Q18-1015

Branchburg 138 kV

January 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1015. The Interconnection Customer proposes a 361.5 MW wind facility interconnecting at the Branchburg 138 kV substation in the New Jersey area. The request was received on June 22, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 333 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,939,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Magnet Cove - Branchburg 138kV
Contingency: Mcleansboro - Branchburg 138kV
Loading: 195.6% (Rating: 187 MVA)
DFAX: 100.0%
Pre-project loading: 1.8%
2. Facility: Mcleansboro - Branchburg 138kV Line
Contingency: Magnet Cove - Branchburg 138kV
Loading: 174.3% (Rating: 209 MVA)
DFAX: 100.0%
Pre-project loading: 1.6%
3. Facility: Mcleansboro - W Vernon 138kV L
Contingency: Maquoketa 345/138kV Transformer Bank
Loading: 100.0% (Rating: 194 MVA)
DFAX: 22.7%
Pre-project loading: 57.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Magnet Cove - Branchburg 138kV	Rebuild line with higher-capacity conductor	\$12,784,000	100% (\$12,784,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Branchburg - Mcleansboro Substation 138kV	Rebuild line with higher-capacity conductor	\$12,981,000	100% (\$12,981,000)
West Vernon - Mcleansboro 138kV	Reconductor line	\$1,174,000	100% (\$1,174,000)
	Total allocated to project		\$26,939,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.